Pre-Analysis Plan (PAP) — ManyDaughters Project

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Dataset = SOEP-Core from 2000 to 2020

Restrictions and coding

- Since the decision to have children is endogenous and may correlate with unobserved factors affecting our outcomes, we restrict our sample to respondents who have at least one child. This restriction helps mitigate selection bias and allows us to focus on variation related to the gender composition of children rather than the decision to become a parent.
- Observations with missing key variables will be excluded.
- Individuals younger than 18 years old will be excluded.
- We will use the original SOEP variables but recode them where necessary, primarily into binary variables, to ensure clarity and consistency in the analysis.
- We will use robust standard errors.
- We will study the period 2000-2020.

Hypotheses Tested

1. ***Having daughters leads to stronger support for gender equality (as measured by attitudes).***
   That is, denoting *X* as a measure of support for gender equality, (*X* for respondents with daughters) > (*X* for respondents without daughters).

Variables and regression

For H1, we estimate the following linear regression model:

$Y_{it} = \beta_0 + \beta_1\ Daughters_{it} + \beta_2\ \boldsymbol{X}_{it}\ + \gamma_t\ + \delta_r + \varepsilon_{it}$

where:

- Y_{it} is the outcome variable for individual *i* at time *t*. We use the attitude regarding the statement "Best if man and woman work the same amount so they can share the responsibility" (plh0308_v2, wave 2018). Respondents answer on a 7-point Likert scale from 1=completely disagree to 7=fully agree. We drop subjects who did not provide an answer (coded as -1)
- $Daughters_{it}$ is the binary indicator for having at least one daughter. Equal to 1 if the respondent has at least one daughter, 0 otherwise.

- $\boldsymbol{X}_{it}$ includes control variables:
  - Respondent gender: Binary indicator of gender equal to 1 if gender is male, zero if female.
  - Respondent age (continuous)
  - Respondent education level (years of schooling/training)
  - Household net income (logged)
  - Employment status (full-time employed/part-time employed, unemployed)
  - Marital status (married, single, etc.)
  - Number of children (continuous)
  - Respondent living in East Germany (binary, 1 if East, 0 if West Germany)

- γ_t are survey year fixed effects, using the wave identifier
- ε_{it} is the error term

Interpretation of results

The significance of the effect will be determined by the p-value associated with the coefficient β_1. A p-value less than 0.05 indicates statistical significance. If the p-value for β_1 is below 0.05, we reject the null hypothesis and conclude that the hypothesis "Having daughters increases support for gender equality attitudes" (compared to people without daughters, i.e., with sons) cannot be rejected.

2. ***Having daughters leads to more left-leaning political preferences.***
   That is, denoting Y as a measure of left-leaning political preferences, (Y for respondents with daughters) > (Y for respondents without daughters).

Variables and regression

For H2, we estimate the following linear regression model:

$Y_{it} = \beta_0 + \beta_1 \, Daughters_{it} + \beta_2 \, \boldsymbol{X}_{it} + \gamma_t + \delta_r + \varepsilon_{it}$

where:

- Y_{it} is the outcome variable for individual *i* at time *t*. The outcome variable is based on the question, "How would you rate your political views? [0] completely left to [10] completely right." We will recode this variable so that higher values correspond to more far-left political views. We drop subjects who did not provide an answer (coded as -1)
- $Daughters_{it}$ is the binary indicator for having at least one daughter. Equal to 1 if the respondent has at least one daughter, 0 otherwise.
- $\boldsymbol{X}_{it}$ includes control variables:
  - Respondent gender: Binary indicator of gender equal to 1 if gender is male, zero if female.
  - Respondent age (continuous)
  - Respondent education level (years of schooling/training)
  - Household net income (logged)
  - Employment status (full-time employed/part-time employed, unemployed)
  - Marital status (married, single, etc.)
  - Number of children (continuous)
  - Respondent living in East Germany (binary, 1 if East, 0 if West Germany)

- γ_t are survey year fixed effects, using the wave identifier
- ε_{it} is the error term

Interpretation of results

The significance of the effect will be determined by the p-value associated with the coefficient β_1. A p-value less than 0.05 indicates statistical significance. If the p-value for β_1 is below 0.05, we reject the null hypothesis and conclude that the hypothesis "Having daughters leads to more left-leaning political preferences." (compared to people without daughters, i.e., with sons) cannot be rejected.

3. ***Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).***

That is, denoting *Z* as a measure of gender equality in the household, (*Z* for respondents with daughters) > (*Z* for respondents without daughters).

For this analysis, we will merge partners using the partner id and focus on the sample where female and male partner are available.

Variables and regression

For H3, we estimate the following linear regression model:
$Y_{it} = \beta_0 + \beta_1 \, Daughters_{it} + \beta_2 \, \boldsymbol{X}_{it} + \gamma_t + \delta_r + \varepsilon_{it}$
where:

- Y_{it} is the outcome variable for individual *i* at time *t*. The outcome variable measures gender equality in the household by capturing the share a respondent spends on housework and childcare compared to their partner. We will sum the hours respondents report spending on housework and childcare during a typical weekday. We will calculate the respondent's share of total household labor. We will subtract 0.5 from this number and take the absolute value.
- $Daughters_{it}$ is the binary indicator for having at least one daughter. Equal to 1 if the respondent has at least one daughter, 0 otherwise.
- $\boldsymbol{X}_{it}$ includes control variables:
  - Respondent gender: Binary indicator of gender equal to 1 if gender is male, zero if female.
  - Respondent age (continuous)
  - Respondent education level (years of schooling/training)
  - Household net income (logged)
  - Employment status (employed/unemployed)
  - Marital status (married, single, etc.)
  - Number of children (continuous)
  - Work hours

- γ_t are survey year fixed effects, using the wave identifier
- δ_r are regional fixed effects, using East-West
- ε_{it} is the error term

Interpretation of results

The significance of the effect will be determined by the p-value associated with the coefficient β_1. A p-value less than 0.05 indicates statistical significance. If the p-value for β_1 is below 0.05, we reject the null hypothesis and conclude that the hypothesis "Having daughters leads to less gender equality in the household." (compared to people without daughters, i.e., with sons) cannot be rejected.

4. ***Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).***
   That is, denoting *X* as a measure of support for gender equality, (*X* for male respondents with daughters - *X* for male respondents without daughters) > (*X* for female respondents with daughters - *X* for female respondents without daughters).

Variables and regression

For H4, we estimate the following linear regression model:

$Y_{it} = \beta_0 + \beta_1\, Daughters_{it} + \beta_2\, Gender_i + \beta_3(Daughters_{it} * Gender_i) + \beta_4\, \boldsymbol{X}_{it} + \gamma_t + \delta_r + \varepsilon_{it}$

where:

- Y_{it} is the outcome variable for individual *i* at time *t*. We use the attitude regarding the statement "Best if man and woman work the same amount so they can share the responsibility" (plh0308_v2, wave 2018). Respondents answer on a 7-point Likert scale from 1=completely disagree to 7=fully agree. We drop subjects who did not provide an answer (coded as -1)
- $Daughters_{it}$ is the binary indicator for having at least one daughter. Equal to 1 if the respondent has at least one daughter, 0 otherwise.
- $Gender_i$ is binary indicator of gender equal to 1 if gender is male, zero if female.
- $Daughters_{it} * Gender_i$ is the interaction term indicating the difference in effect between genders.
- $\boldsymbol{X}_{it}$ includes control variables:

- Respondent age (continuous)
- Respondent education level (years of schooling/training)
- Household net income (logged)
- Employment status (employed/unemployed)
- Marital status (married, single, etc.)
- Number of children (continuous)

- γ_t are survey year fixed effects, using the wave identifier
- ε_{it} is the error term

Interpretation of results

The significance of the effect will be determined by the p-value associated with the coefficient β_3 on the interaction term between having at least one daughter and the respondent's gender. A p-value less than 0.05 indicates statistical significance. If the p-value for β_3 is below 0.05, we reject the null hypothesis and conclude that the effect of having daughters on support for gender equality differs significantly between men and women. Specifically, this supports the hypothesis that "Having daughters leads to a larger increase in support for gender equality for men than for women."